Selena Zheng

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SKILLS & TECHNICAL TOOLS

AI & Developer Tools: Claude, Gemini, Cursor, Prompt Engineering, Git, Docker, Kubernetes, Jupyter Notebooks, Jira
Programming: Python, Java, Scala, JavaScript, SQL, React, HTML/CSS
Creative Technology: Figma, Blender, Creative Coding, FDM 3D Printing, Computational Design, p5.js, ml5.js

WORK EXPERIENCE

WORKDAY **Pleasanton, CA**

Sr. Associate Software Developer Sep 2024 - Present

- Refactored Prism Analytic's proxy infrastructure by deprecating legacy Nginx ModSecurity configuration and reconfiguring ingress port routing, reducing proxy latency by 20% and streamlining network traffic flow
- Restructured the implementation of Apache Zookeeper and Curator to handle Zookeeper connection losses gracefully, eliminating restart loops and reducing tenant registration discrepancies
- Integrated GenAI workflows utilizing Cursor and Claude alongside Grafana and Prometheus to diagnose system bottlenecks, executing prompt-driven debugging to accelerate distributed infrastructure optimizations
- Developed and maintained automated data post-processing script in Jupyter Notebooks, utilizing Trino API and Python to validate over 30,000+ tenant activations per month in Workday's Prism Analytics across production and test environments, cutting manual verification efforts by 90%
- Partnered cross-functionally across multiple technical teams within the Prism Analytics org to diagnose and resolve critical bugs in Java and Scala, maximizing system availability for global enterprise tenants

Software Development Engineering Intern May 2023 – Aug 2023

- Implemented distributed tracing for Prism Analytics server using OpenTelemetry, Jaeger UI, Docker, and Kubernetes
- Owned 10+ Jira tickets, authored POC documentation, and presented final architecture to an engineering audience of 30+ people

NBCUNIVERSAL **New York, NY**

Software Engineering Intern (Peacock) Sep 2022 – Dec 2022

- Bridged the gap between engineering and non-technical business stakeholders by designing and presenting an interactive API documentation visualizer to streamline client handoffs
- Optimized platform reliability for Peacock's Live Event Manager by deploying automated testing frameworks to rapidly resolve technical bugs and JIRA tickets ahead of live broadcast events

Media Tech Intern Jun 2022 – Aug 2022

- Built Python-based machine learning workflows using autoencoders to improve media processing efficiency across Peacock's content infrastructure
- Collaborated cross-functionally within an agile team of 8 to design and launch an interactive, centralized digital media platform for internal stakeholders

PROJECTS

Kinetic Typography **2026**

- Built an end-to-end AI powered kinetic typography generator in React and Typescript that analyzes uploaded audio and uses Google Gemini to generate creative direction, driving a motion engineer with 15+ customizable behavior modules
- Implemented in-browser video export and deployed app on Vercel with a serverless API layer including request validation and rate limiting.

3D Printed Chainmail **2024**

- Designed a procedural computational design pipeline that transforms arbitrary 2D vectors into printable, interlocking 3D chainmail structures through parametric geometry generation through a custom Blender Python add-on
- Iteratively tested PLA and nylon fabrication parameters through rapid FDM prototyping, optimizing print reliability, mechanical performance, and flexibility

Interactive Installation **2023**

- Built an interactive installation using real-time computer vision (ml5.js), React, and p5.js to transform live environmental inputs into dynamic visual experiences
- Managed the end-to-end production pipeline, taking the tech stack from an experimental concept to a stable public installation for an audience of 100+ people

EDUCATION

BOSTON UNIVERSITY **Boston, MA**

Bachelor of Arts, Computer Science